



**Shri Vile Parle Kelavani Mandal's  
USHA PRAVIN GANDHI COLLEGE OF ARTS,  
SCIENCE AND COMMERCE (Autonomous)**



# **AI-RESCUE: INTELLIGENT DISASTER RESPONSE & RESOURCE COORDINATION PLATFORM**

**A Field Project Report  
Submitted in Partial Fulfillment  
For Assessment in  
Third Year  
BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)**

Submitted by

Name: Dev Vikam

SAP ID: 53013240013

**Under the esteemed guidance of  
Dr. Manisha Divate**

## **CERTIFICATE**

This is to certify that Mr. Dev Vikam, SAP ID 53013240013, has successfully completed the Field Project Work in B.Sc. Information Technology under the Department of Information Technology with the project title "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" under the esteemed guidance of Dr. Manisha Divate, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Information Technology from Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), for the Academic Year 2026-2027.

**Dr. Manisha Divate**  
**Project Guide**

**Course Coordinator /**  
**HOD**

**Principal**  
**UPG College**

**Date:** \_\_\_\_\_

**External Examiner**

# **DISASTER RESPONSE & CRISIS TECH LABS**

Center for Disaster Informatics, AI Solutions & Emergency Coordination  
Mumbai, Maharashtra - 400056

## **COMPANY CERTIFICATE**

This is to certify that Mr. Dev Vikam, a student of Third Year B.Sc. Information Technology (SAP ID: 53013240013) from Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), has successfully undergone and completed his Field Project work on the project entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" from June 2026 to September 2026 under the mentorship and technical supervision of our technical engineering team.

During this 90-hour engagement (30 hours guide-supervised reviews and 60 hours independent development work), he demonstrated excellent analytical capabilities, technical proficiency in full-stack architecture, spatial query algorithms, and high commitment to real-time humanitarian engineering solutions. His conduct and performance throughout the tenure were exemplary.

**Date: 26th September 2026**  
**Place: Mumbai**

**Authorized Signatory & Seal**  
**Disaster Response & Crisis Tech Labs**

## **DECLARATION**

I, Mr. Dev Vikam, hereby declare that the project report entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" represents my original work carried out under the guidance of Dr. Manisha Divate.

The work presented in this report has been completed as part of the Third Year Bachelor of Science (Information Technology) program at Shri Vile Parle Kelavani Mandal's Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous) for the academic year 2026-2027.

I further declare that this report has not been submitted previously to any other university, institute, or board for the award of any degree, diploma, fellowship, or other similar academic recognition. All information obtained from published literature, software libraries, and web resources has been duly cited and acknowledged in the bibliography.

**Date: 26th September 2026**  
**Place: Mumbai**

**Mr. Dev Vikam**  
**SAP ID: 53013240013**

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# LOG BOOK

## Field Project – 90 Hours

**Project Title: AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform**

### GUIDE SESSIONS – 30 HOURS

| Date       | Hours   | Activity                                                                                                                                |
|------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 12/06/2026 | 3 hours | Project orientation and AI-Rescue overview; understanding the problem of fragmented disaster response and defining project scope.       |
| 18/06/2026 | 3 hours | Requirement gathering discussion: citizen incident reporting, SOS alerts, responder mobilization, and resource allocation requirements. |
| 20/06/2026 | 3 hours | Review of proposed technology stack: Next.js frontend, FastAPI backend, PostgreSQL/PostGIS spatial database, and Redis caching.         |
| 25/06/2026 | 3 hours | Detailed UI/UX workflow discussion: citizen reporting portal, admin command center, and first responder mobile interfaces.              |
| 27/06/2026 | 3 hours | Resource allocation logic review: matching nearest available resources, hospitals, and shelters to high-severity incidents.             |
| 02/07/2026 | 3 hours | Review of functional and non-functional requirements; performance benchmarks for sub-second SOS alert dispatching.                      |
| 23/07/2026 | 3 hours | Project schedule and milestone planning; Gantt chart construction and Work Breakdown Structure (WBS) approval.                          |
| 13/08/2026 | 3 hours | System architecture, ER Diagram (ERD), Data Flow Diagrams (DFDs), and end-to-end System Flowchart review.                               |
| 29/08/2026 | 3 hours | Relational database schema finalization (16 tables) and AI multi-modal severity scoring module validation.                              |
| 24/09/2026 | 3 hours | Final documentation review, project verification, thesis binding checks, and final viva examination sign-off.                           |

**Total: 30 hours guide-supervised project work**

# LOG BOOK

## Field Project – 90 Hours

**Project Title: AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform**

## WORK – 60 HOURS

| Date       | Hours   | Activity                                                                                                                       |
|------------|---------|--------------------------------------------------------------------------------------------------------------------------------|
| 12/06/2026 | 3 hours | Field study on disaster communication delays; comprehensive requirement analysis for AI-Rescue.                                |
| 13/06/2026 | 3 hours | Literature survey of FEMA, NDMA, Ushahidi, and Sahana disaster response frameworks.                                            |
| 15/06/2026 | 3 hours | Next.js 14 and FastAPI development environment setup; Docker containerization configuration.                                   |
| 17/06/2026 | 3 hours | UI wireframing for Citizen SOS dispatch, Incident Map view, and Resource Tracker dashboards.                                   |
| 19/06/2026 | 3 hours | PostgreSQL database initialization with PostGIS spatial extensions for geo-tagging coordinate storage.                         |
| 20/06/2026 | 3 hours | Authentication module implementation: JWT authentication, bcrypt hashing, and Role-Based Access Control.                       |
| 22/06/2026 | 3 hours | Citizen Incident Reporting module: GPS reverse-geocoding, multi-category selection, and photo upload pipeline.                 |
| 24/06/2026 | 3 hours | Live incident feed integration: WebSocket pub/sub implementation via Redis for real-time map updates.                          |
| 26/06/2026 | 3 hours | AI multi-modal NLP classification pipeline: text severity scoring and OCR text extraction from disaster images.                |
| 27/06/2026 | 3 hours | Haversine & PostGIS spatial query engine implementation for nearest emergency resource discovery.                              |
| 31/07/2026 | 3 hours | Continuous System Flowchart design modeling 41 operational nodes, decision diamonds, and error-retry bus loops.                |
| 06/08/2026 | 3 hours | Multi-modal AI severity scoring algorithm formulation, confidence weighting, and verification against synthetic test datasets. |
| 10/08/2026 | 3 hours | Automated SMS, Email, and Push notification microservice development using Twilio and Firebase Cloud Messaging.                |
| 15/08/2026 | 3 hours | Offline-first capability development using IndexedDB and Service Workers for low-connectivity disaster zones.                  |
| 20/08/2026 | 3 hours | System integration testing: simulating simultaneous mass casualty incident reports and automated responder dispatches.         |
| 25/08/2026 | 3 hours | Security hardening: SQL injection prevention, rate limiting on emergency endpoints, and encrypted audit logging.               |
| 02/09/2026 | 3 hours | ER Diagram and Relational Schema finalization across 16 database entities and referential integrity constraints.               |
| 08/09/2026 | 3 hours | User Interface design polish: Leaflet heatmaps, emergency resource status indicators, and responsive mobile layouts.           |
| 14/09/2026 | 3 hours | Comprehensive test case authoring and execution across unit, integration, stress, and disaster failover scenarios.             |
| 20/09/2026 | 3 hours | Final report proofreading, high-resolution diagram vectorization, logbook synchronization, and submission binding.             |

**Total Student Work: 60 Hours | Combined Field Project Total: 90 Hours Completed**

**Name & Signature of Company / Organization**

**Disaster Response & Crisis Tech Labs**

**Name & Signature of Student**

**Dev Vikam**

**Name & Signature of Teacher**

**Dr. Manisha Divate**